A Literal Source-Profile Angle and Dimension Ladder for Finite Twin-Prime Shells

Liang Wang
School of Mathematics and Statistics
Huazhong University of Science and Technology (HUST)
Wuhan, China

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Abstract

TPC-297 replaced a frozen source ray by four literal cutoff profiles, but left open how much source dimension is needed to approach a physically selected signed target. We order seventeen literal profiles $\beta_z(t) = \lambda(t) - \sum_{d \le z,\ d|t} \mu(d)$ by $z = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61$ and study every prefix. If A is the frozen physical shell matrix, U_k is the first k source profiles, and $V_k = A^{\mathsf{T}} U_k$, we prove that the least-squares residual is the sine of the angle from the target to $\mathrm{col}(V_k)$, and that it decreases along the nested ladder. On the inherited 18-row, 1,380-edge grid, two modular replays certify the expected rank $\min(k, |S|)$ for all 306 tested prefixes. The weighted sign target first reaches normalized RMS 1/2 only after at least two thirds of the shell dimension on every row, while the all-positive control reaches the same threshold in at most six profiles on every row. The final finite prefix spans every registered target space. This is a finite dimension/angle atlas, not an asymptotic native-profile theorem or a twin-prime result.

1 Position on the route

The finite prime-shell line has separated ambient image existence from native source geometry. TPC-294 selected signed targets using the physical Gram magnitudes; TPC-295 showed that unrestricted rational source coordinates can hit every finite target when the shell Gram is full rank; and TPC-296 found that those unrestricted witnesses are cheap while a frozen one-ray proxy is poorly aligned [3, 1]. TPC-297 then tested four literal cutoffs and found a strong positive control but persistent weighted-target separation [2].

The present paper asks the smallest natural continuation: along an ordered literal source family, at what dimension does the target become close? The answer is deliberately comparative. The all-positive target is captured quickly, whereas the magnitude-weighted sign target remains diffuse in the profile image until a large fraction of the shell dimension is present.

2 Literal profiles and the physical map

For an integer t, set

$$\lambda(t) = \begin{cases} 1/j, & t = p^j \text{ for a prime } p, \\ 0, & \text{otherwise}, \end{cases} \qquad \beta_z(t) = \lambda(t) - \sum_{\substack{d \le z \\ d|t}} \mu(d).$$

Let

$$Z = (3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61)$$

and let U_k contain the first k vectors $u_z = (\beta_z(t))_{t \in I}$. The physical columns are frozen from the registered source profile:

$$(g_q)_u = \sum_{\substack{t \in I \\ t \ne u}} q K_{H,s}(u - t) B_q(u, t) \beta_t, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod q} - \frac{1}{q - 1} \right),$$

where $K_{H,s}(v) = H^{2s}/(H^2 + v^2)^s$. For the prime shell $S = \{q : Q < q \le 2Q\}$, write

$$A = [g_q]_{q \in S}, \qquad V_k = A^{\mathsf{T}} U_k \in \mathbb{R}^{S \times k}.$$

The source directions are selected before the target is inspected. The weighted minimum, unit-edge max-cut, and all-positive vectors are only finite target controls in this paper.

Remark 1. *The term literal refers to the finite cutoff formula and its source-side origin. It does not claim that this seventeen-element list is the complete native profile class, nor that its dimension or conditioning is uniform in N.*

3 Projection, angle, and dimension theorem

For a full-column-rank prefix define

$$P_k = V_k(V_k^\mathsf{T} V_k)^{-1}V_k^\mathsf{T}, \qquad c_k = (V_k^\mathsf{T} V_k)^{-1}V_k^\mathsf{T} b.$$

Theorem 1 (Finite profile-angle identity)**.** *For every target $b \in \mathbb{R}^S$ and every full-column-rank prefix,*

$$\min_c \|V_k c - b\|_2^2 = b^\mathsf{T}(I - P_k)b.$$

If θ_k is the angle between the vector b and the subspace $\mathrm{col}(V_k)$*, then*

$$r_k := \frac{\min_c \|V_k c - b\|_2}{\|b\|_2} = \sin\theta_k, \qquad \cos^2\theta_k = \frac{b^\mathsf{T} P_k b}{\|b\|_2^2}.$$

Proof. The vector $P_k b$ is the orthogonal projection of b onto $\mathrm{col}(V_k)$. For any y in that subspace, $P_k b - y$ is orthogonal to $(I - P_k)b$. Pythagoras gives

$$\|b - y\|_2^2 = \|(I - P_k)b\|_2^2 + \|P_k b - y\|_2^2,$$

with equality at $y = P_k b$. This proves the least-squares identity. Dividing the same orthogonal decomposition by $\|b\|_2^2$ gives the two angle relations. □

Proposition 1 (Nested prefixes)**.** *For $k < \ell$,* $\mathrm{col}(V_k) \subseteq \mathrm{col}(V_\ell)$*. Consequently $r_\ell \le r_k$ and $\theta_\ell \le \theta_k$.*

Proof. The columns of U_k are the first columns of U_ℓ, so the same is true after applying the linear map A^T. Distance to a larger subspace cannot increase, and arcsin is increasing on $[0, 1]$. □

For a threshold τ, the finite dimension statistic is

$$k_\tau(b) = \min\{k : r_k \le \tau\}.$$

For a sign target $b \in \{-1, 1\}^{|S|}$, r_k is exactly the RMS residual after division by $\sqrt{|S|}$. If the first $|S|$ columns have rank $|S|$, then $P_{|S|} = I$ on the target space and the finite residual is zero.

4 Finite audit and validation

The audit inherits eight growth rows, four exponent-crossover rows, and six source-control rows from the preceding releases. Each physical and profile entry is formed as a rational number. For every $k = 1, \dots, 17$, Gaussian elimination modulo 1000000007 and 998244353 checks the rank of V_k. For $k \le |S|$, 70-digit QR arithmetic computes residuals, principal-angle sines, captured fractions, coefficient norms, and nonzero singular-value condition numbers. The final tested prefix is $k = |S|$ on every row.

The independent checker reconstructs physical columns source-first and does not import the producer. It recomputes all rank entries, residual intervals, angle identities, monotonicity, and threshold indices. Exact fixtures test a duplicated late column, in-image and out-of-image targets, and the threshold bookkeeping. Normal and optimized interpreter runs must agree with empty standard error.

The minimum weighted dimension fraction is exactly the declared finite floor $2/3$ (attained already on the three-prime row and on several larger rows). The all-positive threshold dimensions range from one to two on the larger representative rows and never exceed six in the full census. At the last prefix, the rank certificate makes all three finite target controls exactly representable in the rational target space; the high-precision residual is a roundoff enclosure around zero.

The largest displayed prefix condition upper bound in the certificate is approximately 4.24×10^4. This number is recorded as a finite conditioning diagnostic, not as a theorem about the moving ladder.

Table 1: TPC-298 finite profile-dimension headline.

quantity	count
literal cutoffs / tested prefixes	17 / 306
rows / shell edges	18 / 1,380
expected prefix-rank ladder (both moduli)	18 / 18
weighted half-RMS dimension ratio $\geq 2/3$	18 / 18
all-positive half-RMS dimension ≤ 6	18 / 18
full-prefix finite target capture	18 / 18
fixed-power credit	0

Table 2: Representative prefix-$k = 4$ residuals and threshold dimensions.

N	H	Q	s	$\lvert S\rvert$	$k_{1/2}^{\mathrm{wt}}$	r_4^{wt}
128	24	9	2	3	2	0
192	32	16	2	5	5	0.7642
256	38	27	2	7	6	0.7051
384	50	40	2	10	8	0.9931
512	58	60	2	13	9	0.9501
512	58	90	2	17	13	0.7853
384	48	70	2	15	13	0.7636
384	52	70	2	15	12	0.8653

5 Interpretation and obstruction

The ladder resolves an ambiguity left by a fixed four-profile snapshot. The positive control is compatible with a low-dimensional literal source image, but the physical weighted sign target is not: half-RMS accuracy requires a large fraction of the available shell dimension on every registered row. This is a geometric statement about the finite map $A^{\mathsf{T}}U_k$.

The result also explains why the final-prefix zero is not a paradox. Once k reaches the shell size and the prefix has full row rank, the restricted finite image is all of $\mathbb{R}^S$, just as TPC-295's unrestricted image was. The meaningful next question is the simultaneous growth of profile dimension, least-norm source cost, and condition number before that trivial finite surjectivity point.

6 Claim boundary and conclusion

The projection, principal-angle, and nested-prefix statements are exact finite linear algebra. The rank ladder and dimension counts are numerically certified finite observations for the declared grid and two moduli. The cutoff list and half-RMS threshold are modeling choices. Nothing here pays arithmetic L^2 credit, a fixed power, or a Gate-B endpoint, and no twin prime conclusion is claimed.

The reusable interface is

$$\text{literal source prefixes } U_k \longrightarrow A^{\mathsf{T}}U_k \longrightarrow P_k \longrightarrow (\text{angle}, \text{dimension}, \text{budget}).$$

The next route test is to compare the weighted dimension ladder with the least-norm source budget and the observed conditioning.

Reproducibility

From the project directory run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc298_profile_angle_dimension_certificate.py --check
python -B experiments/tpc298_independent_checker.py
python -B experiments/tpc298_ladder_stress.py
```

The Session-named Route-A/Route-B evaluator files are absent from this checkout. The theorem ledger, proof package, canonical certificate, independent replay, stress test, PDF audit, and Bridge-B checker are the available fail-closed validation path.

References

[1] Liang Wang. Least-norm source budgets and native-ray obstruction, 2026. TPC-296 project record.

[2] Liang Wang. A literal four-direction source-profile span for finite twin-prime shells, 2026. TPC-297 project record.

[3] Liang Wang. Source-correlation image audit for finite prime shells, 2026. TPC-295 project record.